

```
if __name__ == "__main__":  
    data = [8, 3, 9, 1, 6]  
    try:  
        minimum, maximum = find_min_max(data)  
        print(f"List: {data}")  
        print(f"Minimum value: {minimum}")  
        print(f"Maximum value: {maximum}")  
    except (ValueError, TypeError) as e:  
        print(f"Error: {e}")
```

List: [8, 3, 9, 1, 6]

Minimum value: 1

Maximum value: 9

```
[1]: print("===LIST EXAMPLE ===")
      fruits=["apple","banana","cherry","apple"]
      print("Original List:",fruits)

      ===LIST EXAMPLE ===
      Original List: ['apple', 'banana', 'cherry', 'apple']
```

```
[2]: print("===LIST EXAMPLE ===")
      fruits=["apple","banana","cherry","apple"]
      print("Original List:",fruits)#Add element
      fruits.append("orange")
      #Remove element
      if "banana" in fruits:
          fruits.remove("banana")
      print(" first fruit:",fruits[0])

      ===LIST EXAMPLE ===
      Original List: ['apple', 'banana', 'cherry', 'apple']
      first fruit: apple
```

```
[3]: print("\n===TUPLE EXAMPLE===")
      coordinates=(10.5,20.3,10.5)
      print("Tuple:",coordinates)
      print("First coordinate:",coordinates[0])

      ===TUPLE EXAMPLE===
      Tuple: (10.5, 20.3, 10.5)
      First coordinate: 10.5
```

```
[10]: # Program to find the maximum and minimum in a list
```

```
def find_min_max(numbers):  
    """  
    Returns the minimum and maximum values from a list of numbers.  
    Includes validation to ensure the list is not empty and contains only numbers.  
    """  
    if not numbers:  
        raise ValueError("The list is empty.")  
    if not all(isinstance(x, (int, float)) for x in numbers):  
        raise TypeError("All elements in the list must be integers or floats.")  
  
    # Using built-in functions for efficiency  
    min_val = min(numbers)  
    max_val = max(numbers)  
    return min_val, max_val
```

```
[8]: # Original list
numbers = [1, 2, 3, 4, 5]

try:
    # Remove the first occurrence of 3
    numbers.remove(3)
except ValueError:
    # This will run if 3 is not found in the list
    print("Element 3 not found in the list.")

print("Updated list:", numbers)
```

Updated list: [1, 2, 4, 5]

```
[9]: numbers = [5, 10, 15, 20]

total = 0
for num in numbers:
    total += num

print("Sum:", total)
```

Sum: 50

```
[7]: # Given list
numbers = [10, 20, 30, 40, 50]

# First element
print("First element:", numbers[0])

# Last element
print("Last element:", numbers[-1])
```

First element: 10
Last element: 50

```
[6]: # Initial list
numbers = [1, 2, 3, 4]

# Number to add
num_to_add = 100

# Validate that the value is an integer before adding
if isinstance(num_to_add, int):
    numbers.append(num_to_add) # Adds to the end of the list
    print("Updated list:", numbers)
else:
    print("Error: Only integers can be added.")
```

Updated list: [1, 2, 3, 4, 100]

```
[11]: print("\n===TUPLE EXAMPLE===")
      coordinates=(10.5,20.3,10.5)
      print("Tuple:",coordinates)
      print("First coordinate:",coordinates[-1])
      x,y,z=coordinates
      print(f"unpacked values:x={x},y={y},z={z}")
```

```
===TUPLE EXAMPLE===
Tuple: (10.5, 20.3, 10.5)
First coordinate: 10.5
unpacked values:x=10.5,y=20.3,z=10.5
```

```
[5]: # Create a list of 5 numbers and print all elements.
      numbers = [10, 20, 30, 40, 50]

      # Print all elements
      for num in numbers:
          print(num)
```

```
10
20
30
40
50
```